

Siddharth Ranjan

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Technical Skills

Languages	Java, Python, SQL
Frameworks & Libraries	Spring Boot, Spring Data JPA, Flask, PySpark, LangChain, FAISS
Technologies	Apache Kafka, Redis, REST APIs, MySQL, Git
Cloud	AWS EC2, Microsoft Azure
AI	Generative AI, Google Generative AI API, Vertex AI Studio

Experience

System Engineer C1 – Prime

Jan 2026 – Present

Tata Consultancy Services

- Built a POC for a file-reading service using **Spring Boot**, **AWS EC2**, and **Redis** distributed caching to minimize repetitive disk access.
- Integrated **Apache Kafka** into a data-transformation POC to support event-driven data processing.
- Implemented REST endpoints with input validation, error handling, and basic health checks for backend services.

Research and Development Intern

Aug 2024 – May 2025

Nokia – FN BBN CU-Hardening

- Engineered a web-based visualization tool that automated test-data aggregation and reduced manual reporting effort by **30%**.
- Developed a **Python** automation script to dynamically generate configuration images, reducing manual analysis time by **over 80%**, from 10 minutes to under 1 minute.
- Refactored a core logic component to resolve argument-specific failures and improve robustness across input scenarios.

Projects

Chat with PDF using Gemini [GitHub] [Live]

Feb 2024

- Developed a GenAI-powered document question-answering application capable of indexing and querying multiple PDFs using **LangChain** and **FAISS**.
- Integrated Google's **Generative AI API** to generate contextual responses from retrieved document content.
- Implemented semantic document retrieval to reduce information lookup from minutes to seconds.

Blogging Platform Backend [GitHub]

Jan 2024 – Feb 2024

- Developed a RESTful backend in **Spring Boot** supporting user registration, post publishing, likes, and comments.
- Designed and implemented CRUD APIs with persistent storage and entity relationships using **Spring Data JPA** and **MySQL**.

Research Publication

- **Performance Assessment of Various Machine Learning Algorithms in Recommendation** – Published in IEEE; evaluated recommendation algorithms including SVD, SVD++, and k-NN on the MovieLens dataset, with SVD++ achieving improved MAE and RMSE. [IEEE]

Achievements & Certifications

- **CODATHON**: Led a team to **1st place** in a 36-hour hackathon by developing **Hope Haven**, a marketplace connecting NGOs with donors.
- **DATAQUEST**: Achieved **2nd Runner Up** in an 8+ hour Machine Learning hackathon organized by the Data Science Club at VIT Chennai.
- **Certifications**: **Claude** Certified Developer – Foundations (Anthropic) | Microsoft Certified: Azure Fundamentals (**AZ-900**) | Microsoft Certified: Azure AI Fundamentals (**AI-901**)

Education

Vellore Institute of Technology, Chennai, India

July 2023 – May 2025

Master of Computer Applications; CGPA: 9.42

Patna Science College, Patna, India

June 2019 – Dec. 2022

Bachelor of Computer Applications; CGPA: 8.64